Distributive property: For any three sets A, B and C

(i) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ [Intersection over union]

(ii) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ [Union over intersection]

Example 1.21 If $A = \{0,2,4,6,8\}, B = \{x : x$ is a prime number and $x < 11\}$ and $C = \{x : x \in N$ and $5 \leq x < 9\}$ then verify $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Solution

Given $A = \{0,2,4,6,8\}$, $B = \{2,3,5,7\}$ and $C = \{5,6,7,8\}$

First, we find $B \cap C = \{5,7\}$, $A \cup (B \cap C) = \{0,2,4,5,6,7,8\}$ ... (1)

Next, $A \cup B = \{0,2,3,4,5,6,7,8\}$, $A \cup C = \{0,2,4,5,6,7,8\}$

Then, $(A \cup B) \cap (A \cup C) = \{0,2,4,5,6,7,8\}$ ... (2)

From (1) and (2), it is verified that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Example 1.22 Verify $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ using Venn diagrams.

Solution

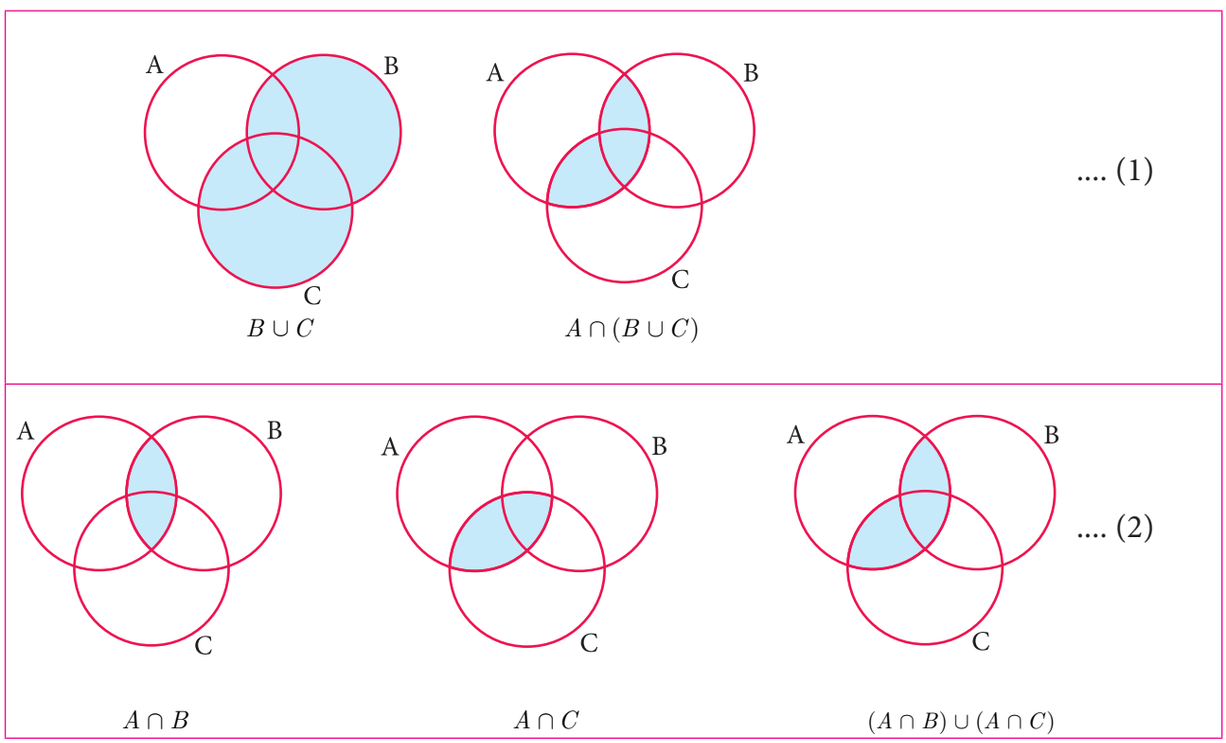


Fig.1.29

From (1) and (2), $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ is verified.

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